Interactive Sinhala Text Reader for the Visually Impaired with the assistance of OCR, NLP, and AI.

R26-DS-002

Project Proposal Report

M.J.J MUTHUNAYAKA – IT22072238

B.Sc. (Hons) Degree in Information Technology Specialized in Data Science

Department of Computer Science

Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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Ms. Adya Dissanayake	19/3/2026	

ABSTRACT

Visual impairment affects approximately 2.2 billion people globally, with around 1.1 million individuals in Sri Lanka experiencing some degree of vision loss [1], [2]. Many depend on printed Sinhala media for daily information, yet no affordable, Sinhala-capable assistive reading solution currently exists.

The identified research gap is twofold: existing assistive tools are predominantly English-centric, and current Sinhala OCR systems fail on heavily stylized headline and display typography — the most information-rich text in Sinhala newspapers — as appearance-based models degrade significantly when font styles deviate from training distributions.

To address this, the project proposes an end-to-end glasses-based, hands-free, voice-interactive Sinhala reading system built on four components: (1) a Topological Sinhala Character Recognition via Medial Axis Transform, (2) a Context-Aware Post-OCR Error Correction Model, (3) an Advanced Document Retrieval and Information Generation module, and (4) an Adaptive Conversational Personalization and Voice Interaction module. This proposal focuses specifically on Component 1, which applies the Medial Axis Transform to skeletonize Sinhala character images into font-invariant structural representations, processed through five stages: image extraction and correction, layout and noise isolation via Otsu Thresholding and Connected Component Analysis, MAT-based skeletonization, tortuosity-based noise mesh rejection, and character recognition using a CNN classification model alongside an exploratory structural route matching approach. The system is implemented in Python using OpenCV, scikit-image, and TensorFlow/PyTorch.

We expect a measurable improvement in stylized Sinhala headline recognition accuracy over the Tesseract/other OCR baseline, delivered as an open-source component compatible with other existing Sinhala OCR and NLP pipelines, as a sort of fallback method. Commercially, the component targets both visually impaired individuals and OCR developers seeking to add font-invariant Sinhala recognition capability. Socially, it directly advances SDG 10 (Reduced Inequalities) and SDG 4 (Quality Education) by enabling equitable, affordable access to Sinhala printed information for the visually impaired community in Sri Lanka.

Keywords: Sinhala OCR, Medial Axis Transform, Topological Skeletonization, Assistive Technology, Font-Invariant Recognition

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I would also like to acknowledge the Sri Lanka Institute of Information Technology for providing the academic environment, and research infrastructure that made this work possible.

Finally, I want to thank my fellow group members, who contributed to the structure and direction of the overall system. While the group collaboratively determined the overall scope and component outlines, each member independently invested to develop and refine their respective components. This ensured both a balanced distribution of work and a high degree of specialization for each component.

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LIST OF ABBREVIATIONS

Abbreviation	**Full Form**
MAT	Medial Axis Transform
OCR	Optical Character Recognition
NLP	Natural Language Processing
TTS	Text-to-Speech
STT	Speech-to-Text
RAG	Retrieval-Augmented Generation
CCA	Connected Component Analysis
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
SDG	Sustainable Development Goal
WER	Word Error Rate
CER	Character Error Rate

1. INTRODUCTION

1.1 Background and Context

Visual impairment is a significant health concern with approximately 2.2 billion people worldwide suffering from some form of vision impairment, with a larger portion of these people residing in low/middle-income countries [1]. In Sri Lanka, it is estimated that around 1.1 million individuals are affected by visual impairment to a certain degree of [2][2]. Many of these individuals are therefore restricted to information from sources such as Newspapers, Textbooks, articles and other various documents.

Sinhala, which is one of the official and majority language of Sri Lanka, is spoken by approximately 17 million people and is written in a rather complex writing system comprising over 50 base characters, numerous diacritics, conjunct letters, and a large set of typographic variations. The inability to access Sinhala printed content translates directly into social, educational, and informational exclusion for visually impaired individuals.

There are many assistive reading technologies available, but they all seem to be predominantly English-centric, like tools such as JAWS, NVDA, and Google Lookout provide support mainly for Latin-script text and offer limited to no functionality for Sinhala script. Existing Sinhala OCR tools, such as those built on Tesseract or other proprietary models, focus mainly on standard body text in uniform fonts and fail significantly when encountering the heavily stylized, decorative, and large-format fonts used in Sinhala newspaper headlines, titles, and topic labels, which tend to carry the most contextual and informational weight. High-end wearable solutions like OrCam or Envision Glasses exist but are out of reach for Sri Lankan users, as they often exceeding $2,000, and also lack sufficient support for the Sinhala language.

We propose a glasses-based, hands-free, voice-interactive reading system tailored for visually impaired Sinhala Text readers. This integrated assistive reading system achieve this through a four-component pipeline: (1) a Topological Sinhala Character Recognition via Medial Axis Transform, (2) a Context-Aware Post-OCR Error Correction Model, (3) an Advanced Document Retrieval and Information Generation module, and (4) an Adaptive Conversational Personalization and Voice Interaction module.

This proposal focuses specifically on Component 1, the Topological Sinhala Character Recognition via Medial Axis Transform which is responsible for accurately and efficiently extracting text from stylized, non-standard Sinhala headline typography, and serves as the critical front-end of the pipeline.

This project aligns with United Nations Sustainable Development Goal 10 (Reduced Inequalities) and SDG 4 (Quality Education), by promoting access to information for persons with disabilities[3]. Furthermore, it aligns with SLIIT's research cluster in Autonomous Intelligent Machines and Systems [4].

1.2 Literature Review

When it comes to the subject of a Sinhala text reader which makes use of similar technologies, in the present work's application domain, Madhusha et al. [8] proposed a mobile-based Sinhala book reader for visually impaired students that combines OCR with text-to-speech output, demonstrating the practical viability of Sinhala OCR in assistive reading applications. However, the system primarily targets printed book body text and does not address the font-invariance challenges present in stylized Sinhala documents. Furthermore, the system relies on basic OCR and a fixed reading style with one-directional text-to-speech interaction. In contrast, the proposed system incorporates a transformer-based post-OCR correction model, RAG-based knowledge retrieval, a two-way conversational interface, and user preference learning to provide a more adaptive and interactive reading experience.

The following reviews focus on relevant scholarly sources based on assistive reading systems, Sinhala OCR, skeletonization and MAT techniques, topological and structural character recognition, as well as document image preprocessing. All sources were verified and critically evaluated for their contribution and as well as being used to discover limitations related to the present project.

Sinhala Language OCR and NLP Research

Existing research on Sinhala-specific OCR and natural language processing forms the most direct foundation for this work.

N. De Silva [5] conducted a comprehensive survey of publicly available Sinhala NLP tools and research, documenting the fragmented state of the ecosystem and confirming that the majority of published tools target standard body text, with no systematic work addressing display or headline typography.

Building on this landscape, Jayasundara et al. [6] developed a deep learning-based Sinhala OCR system integrating enhanced mechanisms into the Tesseract engine, achieving notable accuracy improvements on uniform body text fonts but explicitly acknowledging degraded performance on stylized and decorative fonts due to training distribution mismatch.

A more recent benchmarking study from the University of Moratuwa [7] evaluated multiple OCR engines — including Surya-OCR, Tr-OCR, EasyOCR, and Tesseract — against Sinhala and Tamil document corpora using CER, WER, BLEU, and edit distance metrics, with Surya-OCR achieving the best Sinhala result at 2.61% WER on standard text; critically, none of the evaluated engines were tested against stylized headline fonts, leaving this category unaddressed.

Medial Axis Transform and Skeletonization Techniques

Research on skeletonization and the Medial Axis Transform provides the core theoretical basis for the structural approach proposed in this component.

Lam, Lee, and Suen [9] surveyed twenty thinning algorithms, evaluating connectivity preservation, noise sensitivity, and efficiency. They found skeleton quality is highly sensitive to boundary noise, producing spurious branches and distortions, and no single algorithm performs optimally across all cases. This motivates the use of a Tortuosity Filter in Stage D as a post-skeletonization stabilization and noise-rejection step.

Suresh, Vijaya Kumar, and Ramakrishnan [10] proposed a medial-axis-based thinning strategy specifically designed for printed and handwritten character images, producing one-pixel-width skeletons with reduced spurious branches demonstrating that MAT-based thinning can yield clean, stroke-segmented character representations.

OCR Engines and Structural Document Recognition

OCR and document recognition literature provide both baseline references and structural recognition concepts relevant to this component.

Smith [12] introduced the Tesseract OCR engine, which has since become the most widely adopted open-source OCR baseline and is used as the primary comparison benchmark in this work; Tesseract employs an adaptive classifier trained on appearance-based features, making it inherently susceptible to font distribution shifts of the kind found in stylized Sinhala headlines.

Cheng et al. [11] proposed SSR-GNNs, a stroke-based sketch representation using Graph Neural Networks in which strokes are treated as nodes and the spatial relationships are represented as edges focusing on structural relationships rather than physical pixel appearance. The principle underlying the Path Descriptor and Skeleton Route matching approach proposed in this component — and confirms that moving beyond pixel-level features toward topological structures yields measurable improvements in recognition of robustness.

Assistive Technology for Visually Impaired Users

Research on assistive reading systems for visually impaired users establishes both the real-world need and the standard system architecture that the present work targets.

Prabha et al. [13] proposed Vivoice, a reading assistant for blind users combining OCR with text-to-speech output, demonstrating the practicality of the OCR-to-TTS pipeline for assistive

applications; however, the system targets English text exclusively and makes no provision for complex non-Latin scripts such as Sinhala and no focus stylized typography, a gap that directly motivates the present work. Ionescu et al. It also shows that the visually impaired community strongly prefers hands-free, voice-interactive systems — both design requirements that the broader four-component pipeline of which this work forms Component 1 is specifically engineered to satisfy problems identified in [13].

Existing Research Approaches in OCR and Character Recognition

The research can be broadly divided into two approaches: appearance-based and structural or topological based.

Appearance-based methods dominate modern OCR, using models such as CNNs, LSTMs, and Transformers to map pixel representations directly to character labels. Studies by Jayasundara et al. [6] as well as the benchmark by the University of Moratuwa [7] apply this approach, achieving high accuracy on uniform text. However, the reliance on visual features of the characters makes them sensitive to stylistic variations, causing recognition accuracy to degrade heavily on stylized Sinhala headline fonts, as training datasets are limited to fonts.

Structural or topological approaches represent characters by their geometric construction rather than appearance. Lam, Lee, and Suen [9] showed that thinning and skeletonization can produce connectivity-preserving structural representations less sensitive to stroke variation. Suresh et al. [10] further demonstrated that medial-axis thinning generates clean skeletal structures suitable for comparison. Despite this potential, most OCR research, including Sinhala OCR still rely on appearance-based methods, leaving structural approaches underexplored.

Existing Products, Services, and Their Limitations

A review of existing commercial and open-source products shows major gaps in support for Sinhala, especially heavily stylized fonts. Tesseract OCR [12], the widely used open-source baseline, supports Sinhala and has been used in research such as Jayasundara et al. [6]. However, its appearance-based architecture depends on training fonts, performing well on standard printed text but poorly on decorative Sinhala headlines, as confirmed by the University of Moratuwa benchmark [7]. Surya-OCR showed better results on standard Sinhala text in the same study but has not been evaluated on stylized typography and requires higher computational resources.

Products such as Envision Glasses, Optelec ClearReader, and Google Lookout provide real-time OCR and voice output in the form factor of a glass, but none support Sinhala and are limited to standard body text when reading. The SLIIT Sinhala book reader [8]combines Sinhala OCR with text-to-speech yet focuses on printed book text and does not address font-invariant recognition and acts as a basic text to speech application from that point onwards. No existing product specifically handles font-invariant Sinhala headline recognition, the gap addressed by this work.

1.3 Research Gap

The component addresses a gap in assistive technology for Sinhala-speaking visually impaired users, which is an absence of a refined end-to-end, voice-interactive intelligent reading system that handles the complexity of Sinhala content.

Existing tools either have poor Sinhala support and mostly focus on body text (mainly recognizing the box of text before the individual characters in it, meaning if the text box isn't recognized, the characters will be ignored as well.). They also don’t address stylized typography, adaptive explanation, or personalized voice interaction.

For the Topological Sinhala Character Recognition via Medial Axis Transform, three distinct research gaps are identified:

1. Absence of font-invariant OCR for Sinhala display typography: While body-text Sinhala OCR exists, no published system specifically targets the recognition of heavily stylized headline fonts found in Sinhala newspapers, articles, textbooks, and other documents. While the number of characters in these are low, they hold significant value and offer a strong context to all text that follows. Standard CNN and Transformer-based OCR models degrade significantly when stroke thickness, curvature, and spacing deviate from training data.
2. Underutilization of structural/topological methods for Sinhala script: The Medial Axis Transform and skeleton-based character analysis have been largely abandoned in modern OCR pipelines in favor of appearance-based deep learning. No published work has systematically repositioned MAT as a primary recognition mechanism for the Sinhala script, particularly for its complex conjunct characters and diacritics.
3. Lack of non-body-text focus in Sinhala OCR research: The majority of Sinhala OCR research targets uniform body text. Headlines, titles, and display text which are typically the first content a reader encounters and carry the highest semantic density, remain a largely unaddressed category.

This work addresses these gaps through this component, utilizing topological skeletonization as a font-invariant structural representation of Sinhala characters, enabling robust recognition of stylized headline text that is currently unrecognized by standard OCR systems.

2. OBJECTIVE

2.1 Main Objective

To design, implement, and evaluate a Topological Sinhala Character Recognition via Medial Axis Transform that applies Medial Axis Transform-based skeletonization to accurately recognize stylized large-font Sinhala headline and display text, achieving a measurable improvement in character recognition accuracy as well as speed over standard OCR baselines.

- Specific – Targets stylized Sinhala headline & topic text only
- Measurable - Evaluate Character Error Rate (CER), Word Error Rate (WER),Speed
- Achievable - Uses MAT with either Appearance or Structure based recognition.
- Relevant - Directly addresses a documented OCR gap.
- Time-bound - Completed within the project semester schedule.

2.2 Specific Objectives

Objective 1: To develop a skeletonization pipeline that reduces Sinhala character images to topological 1-pixel-wide skeletal representations using the Medial Axis Transform, preserving structural features such as endpoints and junctions.

Objective 2: To implement a noise isolation strategy using Thresholding and Connected Component Analysis (CCA) that to the best extent, separates headline text from background (large body texts) and retains diacritical marks within a defined proximity threshold.

Objective 3: To apply a Filter for noise mesh rejection that distinguishes valid Sinhala strokes from noise artifacts based on path complexity metrics.

Objective 4: To design and implement a recognition model that classifies Sinhala character from the MAT-derived skeletonized structures, with the use of CNN and LSTM architecture to capture both spatial stroke patterns and structural features.

Objective 5: To investigate the feasibility of a secondary Skeleton Route/Direction Matching alternative structural recognition approach, analyzing travel paths and directional changes approach as a more topologically robust classification method for future development.

Objective 6: To evaluate the complete pipeline against a baseline standard OCR system (e.g., Tesseract) on a dataset of stylized Sinhala headline images, using Character Error Rate (CER), Word Error Rate (WER) and Processing speed.

3. METHODOLOGY

This research adopts a Design Science Research (DSR) methodology, which is appropriate for projects that involve the construction and evaluation of a novel artefact — in this case, the Topological Sinhala Character Recognition via Medial Axis Transform Component.

The DSR process follows three core cycles:

1. **Relevance Cycle** (grounding the problem in real-world OCR failure on Sinhala headlines)
2. **Design Cycle** (iterative design and refinement of the procedural pipeline)
3. **Rigour Cycle** (grounding design decisions in established theory — MAT, topological graph analysis, and noise rejection techniques).

The research proceeds through the following phases:

(1) Problem analysis and literature reviews with Initial documentation, (2) Pipeline & architecture design and prototyping, (3) Dataset preparation (4) Implementation and iterative refinement, and (5) Evaluation and benchmarking against baseline OCR systems.

Technical methods and algorithms

Layout and Noise Isolation

Input images containing Sinhala headline text are first converted to grayscale and binarized using Otsu Thresholding (or alternatives), which determines an optimal intensity threshold for foreground–background separation.

Connected Component Analysis (CCA) is then applied to identify distinct pixel clusters. Components are categorized as Anchor Components, representing the main bodies of characters, and Proximity-Kept Components, representing smaller diacritical marks are retained only if they fall within a predefined Euclidean distance This process isolates the headline text while preserving structurally relevant character elements.

Topological Mapping (Skeletonization)

The cleaned binary images are transformed into structural representations using the Medial Axis Transform (MAT), which reduces each character to a one-pixel-wide skeletal structure derived from the distance transform of the binary image. This operation removes stroke-width variation and stylistic features while preserving the topological structure of the character.

From the resulting skeleton, key structural features such as endpoints (stroke terminations) and junctions (points where multiple strokes intersect) are identified for subsequent analysis.

Noise Mesh Rejection (Path Complexity Filtering)

Artifacts produced during thresholding & skeletonization are removed using a path complexity filter based on tortuosity, defined as the ratio between the actual path length and the straight-line distance between its endpoints. Valid Sinhala strokes typically exhibit smooth curvature, whereas noise artifacts often contain irregular branches and excessive angular turns. Path segments exceeding a defined tortuosity threshold are therefore discarded, ensuring that only structurally meaningful skeletal paths remain.

Skeleton-Based Character Recognition using CNN–LSTM

The filtered skeleton representations are used as inputs to a hybrid CNN–LSTM recognition model. A Convolutional Neural Network (CNN) extracts spatial features from the skeleton images, capturing local stroke configurations and geometric patterns.

The extracted features are then processed by a Long Short-Term Memory (LSTM) network, which models sequential relationships within the skeleton structure. This architecture enables the system to recognize Sinhala characters by learning both spatial stroke patterns and structural dependencies between skeletal segments.

Skeleton Route and Direction Matching (Exploratory Method)

An alternative structural recognition strategy is also explored through Skeleton Route and Direction Matching. In this approach, the skeleton is interpreted as a traversable route, where paths between endpoints and junctions are analyzed to record directional traveling sequences. These directional sequences form structural signatures that can be compared against reference patterns for character identification. This exploratory method evaluates the feasibility of rule-based topological matching as a robust recognition strategy for highly stylized Sinhala fonts.

Tools and Platforms with justification

Table 1:Tools & Technologies

Tools/Library	Purpose	Justification
Python	Core implementation Language	The industry standard for ML/DA with extensive libraries.
OpenCV	Image Preprocessing, Thresholding	The standard when it comes to computer vision
Scikit-Image	MAT, Distance transform, Skeletonization	Provide, skeletonizing and medial axis function directly.

NumPy / SciPy	Numerical computation, graph traversal	Efficient array operations for skeleton graph analysis
TensorFlow / PyTorch	Classification model (Phase, Classical Approach)	To be determined based on classification approach selected.
Tesseract OCR	Baseline comparison system	Open-source; widely used Sinhala OCR baseline as well as to extract regular text when it comes to whole application
Jupyter Notebook	Prototyping and visualization	Enables iterative development and visual debugging of pipeline stages
Other tools are to be declared as required in future		

Data collection and ethical considerations

Three sources of Datasets will be used in the project.

1. A per character dataset made with the use of publicly available .ttf files, which are to be generated and classified with python and other tools
2. A digital print media library of text from various sources such as e-newspapers, online article prints, government provide textbook pdfs and more.
3. A manually captured document set consisting of Sinhala document photographs taken at realistic angles to replicate the inputs from an eye-level mounted camera.

Ethical considerations to be included :

- Use of publicly available or openly licensed newspaper content only
- No collection of personally identifiable information
- Dataset to be made openly available to support future Sinhala NLP/OCR research. Ethical clearance will be sought from publishers of the printed media if required.

Validation strategy and performance metrics

The following metrics will be used to evaluate the component:

- Character Error Rate (CER): Measures character-level recognition accuracy against labels.
- Word Error Rate (WER): Measures word-level accuracy, accounting for segmentation and recognition errors.

- Precision/Recall on skeleton graph nodes: Evaluates the accuracy of endpoint and junction detection.
- Speed: Individually test out tools by using headline only testing images and comparing speeds on identified images

4. HIGH-LEVEL SYSTEM ARCHITECTURE

Overview of proposed solution and its components

The proposed system is an end-to-end, glasses-based assistive reading platform designed specifically for visually impaired Sinhala newspaper readers. A person wearing the glasses captures an image of printed Sinhala text hands-free. The image's text data is extracted and then processed through four tightly integrated, specialized components that collectively handle everything from raw image input to intelligent, personalized voice output. Component 1 extracts stylized headline text using topological skeletonization, while Component 2 corrects errors in standard body text using a fine-tuned transformer model and when Component 1 provides the extracted headline text, it will take that into account. The extracted text is sent to Component 3 which retrieves contextual knowledge from multiple sources and generates a response using a RAG-powered Large Language Model, with the output format based on the request received from Component 4. Finally, Component 4 interprets the user's spoken intent from the STT and then personalizes the response using stored user preferences and interaction history and delivers the final output as natural Sinhala speech. The pipeline is bidirectional at the voice layer, allowing the user to issue spoken commands at any point, triggering requested continuously between Components 3 and 4 in real time.

Component 1: Topological Sinhala Character Recognition via Medial Axis Transform

- Captures and corrects the input image, isolates headline text using thresholding and CCA, skeletonizes characters via MAT, and classifies them through a CNN model and a Structural Recognition Model, before applying rule-based correction to produce the final headline text output.

Component 2: Context-Aware Post-OCR Error Correction

- Standard body text is extracted via a baseline OCR model. It then preprocesses and corrects OCR-induced errors using a Transformer model fine-tuned on synthetic noisy-clean Sinhala pairs, producing a context-aware corrected Sinhala text output.

Component 3: Advanced Document Retrieval and Information Generation

- Retrieves relevant content from local databases, web pages, and previous scans, converts it to vector embeddings for semantic search, and passes the retrieved context to a Main LLM with cross-session memory to generate simplified or elaborated responses.

Component 4: Adaptive Conversational Personalization and Voice Interaction

- Transcribes user speech, detects intent via a Mini LLM, and applies a personalization layer using stored user preferences and interaction behavior analysis before generating and delivering a tailored Sinhala voice response via Text-To-Speech.

System Diagram of Components

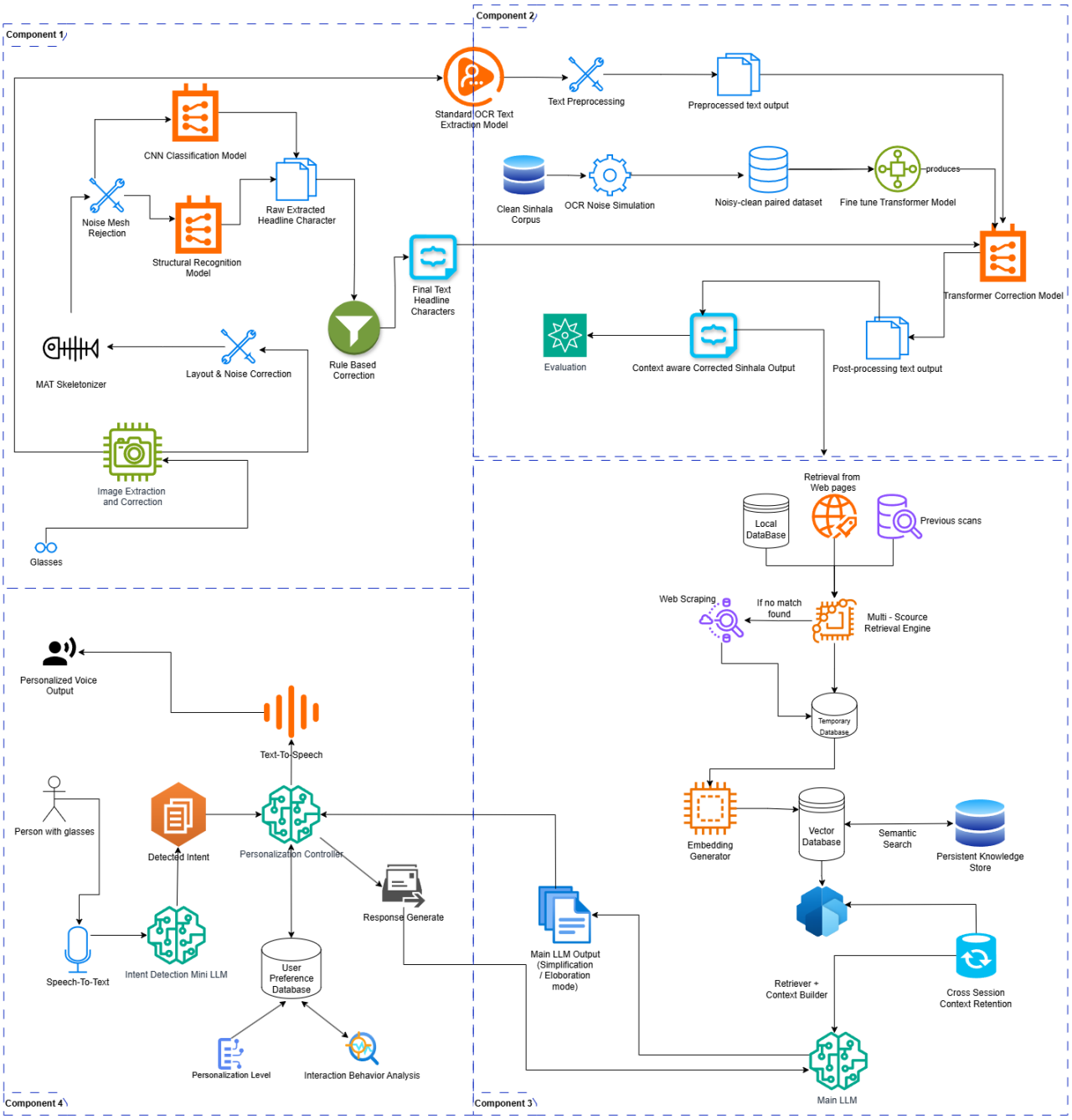


Figure 1:Overall System Architecture

Individual Component

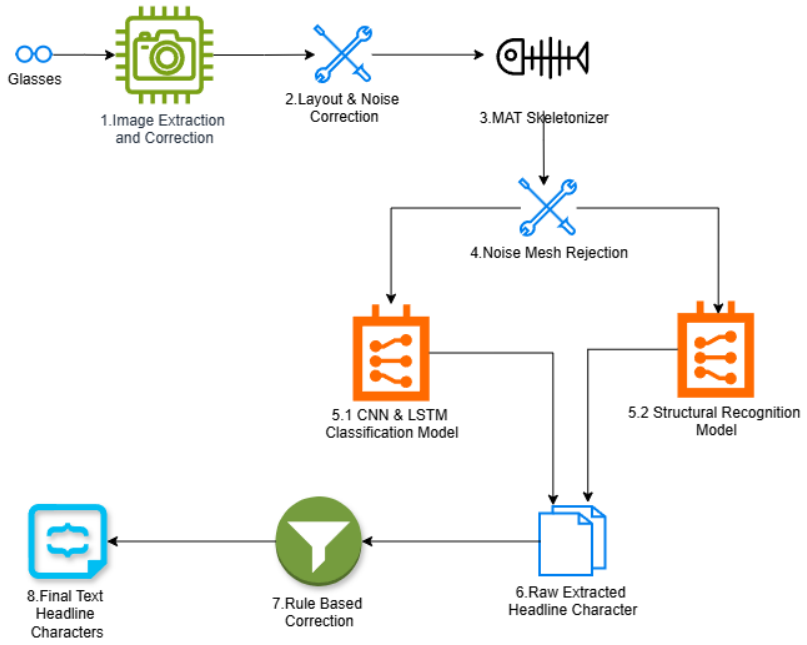


Figure 2: Component Architecture

1.Image Extraction and Corrections – The camera mounted on the glasses captures their image, adding corrections such as skewing to compensate for the angle and any brightness changes needed.

2.Layout & Noise Correction – Perform grayscale conversion and Otsu Thresholding, after which it runs a Connected Component Analysis (CCA) which Anchors component, find proximity components to be kept (such as diatrics, while the rest is treated as noise and discarded.

3.MAT Skeletonizer – Medial Axis Transform (MAT) is applied to the cleaned character regions. The skeletonization process reduces each character to a 1-pixel-wide skeletal structure while preserving its geometric shape. From this skeleton, endpoints and junction points are detected, which represent key structural points of Sinhala characters and prepare the data for structural analysis in later stages.

4.Noise Mesh Rejection – The generated skeleton paths are evaluated using a tortuosity filter, which measures how irregular or twisted a path is. Skeleton segments with high tortuosity, typically caused by residual noise or artifacts, are removed. Only clean and structurally meaningful paths are retained.

5.Character Recognition Models – The cleaned skeleton representation is processed by one of two parallel recognition approaches. The first model uses CNN & LSTM to extract features and classify the Sinhala characters. Alternatively, we could use an exploratory structural recognition model that traverses the skeleton as a directed path, extracting

directional stroke sequences and matching them. Both models generate predicted character outputs with associated confidence values.

6.Raw Extracted Headline Characters – The outputs produced where each predicted character is accompanied by a confidence score, representing the certainty. At this stage, the output still contains potential misclassifications and therefore acts as an intermediate result before correction.

7.Rule-Based Correction – A rule-based standard correction layer is applied to the raw recognized characters. This stage addresses common misclassifications in Sinhala characters, such as confusion between visually similar characters or misplaced diacritics. The rules refine the recognition output to improve textual accuracy before producing the result.

8.Final Text Headline Characters – After rule-based correction, the system produces the final cleaned Sinhala headline text. This corrected output represents the recognized characters extracted from the documents headline & titles and is then passed to the next component of the system, namely the Post-OCR Error Correction module, for further contextual refinement.

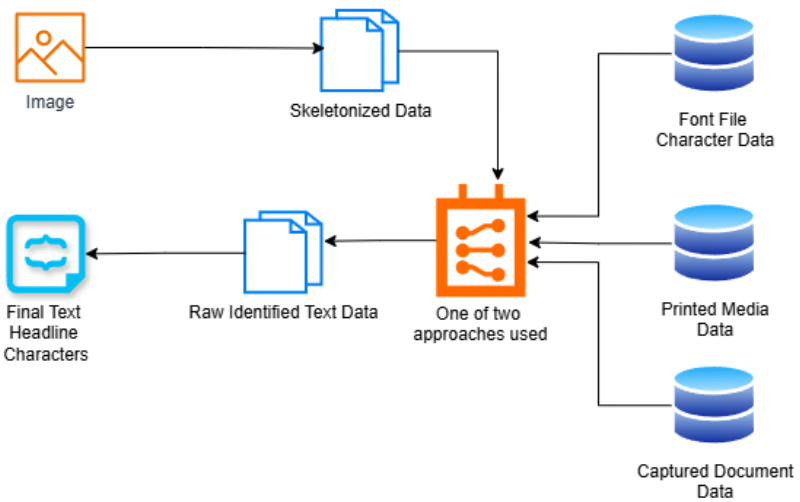
Component Interactions

All Components interact with at least one other component;

- Component 1 and Component 2 run in parallel, where Component 1 handles headlines and finishes first while Component 2 handles body text and when Component 1 output is available, uses it as a high value context for its transformer-based correction.
- Component 3 receives combined corrected text from Component 2 and serves as Main LLM responsible for generation of content
- Component 4 wraps around Component 3, personalizing and voicing its output

The user interacts with Component 4 bidirectionally, receiving output and issuing voice commands, while the camera feeds input to Component1 and 2.

Data flow explanation



The system receives an input image, converts it into skeletonized data, which is then classified by one of two recognition approaches trained on Font File Character Data, Printed Media Data, and Captured Document Data, producing Raw Identified Text Data that is refined into Final Text Headline Characters.

Scalability and security considerations

The MAT-based approach is intentionally computationally lean for the headline-only scope, but it is to be noted that a standard OCR will be running to provide body text which is computationally heavy. Scalability to larger document types or additional scripts is achievable by extending the reference skeleton database, which may be updated with latest font types easily or other types of printed or captured media.

As the system processes only locally captured images without transmitting personal visual data to external servers, privacy and security risks are inherently low

5. USER REQUIREMENTS

Stakeholder identification

Primary Users: Consists of elderly users and persons with acquired vision loss who yearn to read Sinhala text with intent of learning information.

Secondary Users: Developers and researchers integrating the topological OCR module into broader OCR pipelines or assistive tools.

Tertiary Stakeholders: Sri Lankan disability advocacy organizations, open-source NLP community, Sri Lankan organizations in relation to Education, Technology and Literature.

Functional requirements

1. The system shall accept image input containing Sinhala headline text in varying font styles and sizes

2. The system shall perform binarization using Otsu Thresholding to isolate text from the background.

3. The system shall classify connected components into anchor bodies and proximity-kept diacritical marks.

4. The system shall produce a 1-pixel-wide skeletal representation of each character using MAT.

5. The system shall extract and encode path descriptors (DNA sequences) for each skeletal stroke.

6. The system should output a recognized text string for each processed headline image.

7. The system shall provide a confidence score with each recognized output for downstream use by Component 2.

Non-Functional requirements

Performance: The pipeline shall process a single headline image faster than other conventional OCR models on target hardware.

Accuracy: The system shall achieve a CER improvement over the Tesseract/Surya baseline on the test dataset when it comes to headlines.

Usability: The module should output text in a standardized format with relevant information such as confidence score and both corrected and raw text if needed.

Maintainability: Code shall be modular, documented, and open-source, allowing future researchers to extend the reference skeleton database.

Scalability: The reference skeleton database shall be extensible to support additional font styles without retraining the full pipeline.

Reliability: The noise rejection module shall achieve an acceptable false-rejection rate on valid Sinhala strokes.

Commercializing Plan

Target market and customer persona

The primary target users are visually impaired Sinhala-speaking individuals in Sri Lanka who consume newspapers as a source of daily news and information. This includes elderly users with age-related vision loss, individuals with congenital visual impairment, and persons with acquired vision disabilities.

The secondary market comprises developers, OCR tool vendors, and research institutions seeking to integrate a font-invariant Sinhala headline recognition module into existing applications.

Value proposition

It provides a font-invariant OCR capability for Sinhala display typography which is a gap that no existing commercial or open-source tool currently addresses. Furthermore, with a stable implementation of the structural (topology-based) approach, it will be inherently more robust to the font diversity found in real-world Sinhala text.

Revenue model

It will be released as a open-source software module under an appropriate open-source license (e.g., MIT or Apache 2.0). This might help interested parties like the Sinhala NLP research community to easily adopt this technology. The license will benefit assistive technology ecosystems and ensure the widest possible benefit to visually impaired end-users without financial barriers. This would also help devolop this component to a greater extent.

Cost estimation

Since this component relies on the input image, the largest impactable cost would be the camera itself, requiring a higher quality Auto focus industrial level webcam so that it delivers the best quality with a smaller size.

Pricing strategy

The component itself will be given open source with no pricing strategy in mind but other source material required such as detailed architecture, approaches taken and other documentation could be provided at a price along with commercial right if any product want to integrate with it.

Competitive advantage

- Only known system targeting font-invariant recognition of Sinhala typography specifically.
- Structural (topology-based) approach is inherently more generalizable across unseen fonts than trained appearance-based models.
- Open-source release ensures adoption and community-driven improvement.
- Designed as a modular, plug-in & fallback component compatible with existing OCR pipeline

Intellectual property considerations

As an open-source release, no patent protection is sought for this component. The novel contribution lies in the application of MAT-based topological skeletonization to Sinhala display typography and the Path Descriptor (DNA Sequence) encoding scheme, which will be documented in academic publications. Attribution requirements will be enforced through the chosen open-source license.

7. BUDGET AND JUSTIFICATION

Table 2:Budget and Justification

Category	Item	Estimated Cost	Justification
Hardware (External)	Camera, Microphone	LKR 7,000 upwards	These are essential to build the system and even if we use pre-shot images, the better camera directly improves the project results.
Software	All tools open-source (Python, OpenCV, scikit-image)	-None	All primary tools are freely available
Data	.ttf files, Print media from website, capture images	-Depends, as documents to capture images can be bought or borrowed.	Ground truth dataset creation
Man-Hour	Student research time	-Not applicable	Already accounted for in the project
Hardware (Internal)	Computer	-Not applicable	Current available laptops will most be able to the software.

8. WORK BREAKDOWN STRUCTURE (WBS)

Task breakdown per member

Each member will be focused on creating the appliable component since components 3 & 4 heavily rely on results of component 1 & 2. The members are to use a standard “expected” output at each interlinking of components so that each member knows what target output they are needed to create as well as what input they are to expect.

Table 3:Work Breakdown

Member	Component	Expected Input to start with	Expect output from Task
IT22072238	Topological Sinhala Character Recognition via Medial Axis Transform	Images of Sinhala documents	The headlines, both raw and filtered
IT22259134	Context-Aware Sinhala post-OCR error correction model	Images of Sinhala documents (with text extracted from standard OCR model)	Extracted text from the images of the Sinhala document.
IT22120670	Advanced Document retrieval and Information generation	Extracted text from the images of the Sinhala document.	A clear and focused output as per users need based on the scanned document plus additional sources
IT22004468	Adaptive Conversational Personalization and Voice Interaction Module	User’s voice commands as well as generated data from 3rd components.	Data requests that reflect users need and intent as well as TTS that is personalized to user.

Timeline alignment

We plan to keep updated target goals for each stage of development in each component, where each stage of the respective component must be complete at a specific targeted data to ensure the timeline of the components aligned.

We will be making use of a Gantt chart to keep track of the project timeline, with extensive member-wise review done regularly to deeply learn each other's progress & situation

Balanced workload distribution

Each member is entitled to dive deeper into their individual components, as long as the base level is met which is the Expected output from their component. The base level for all components is balanced so that each member gets a fair share of work and content allocated into their component to work with.

Furthermore, in Component 1 & 2, baseline work such as refining and processing the image before use for any of the component is equally distributed.

Integration of components is to be handled by the members on both ends, with the workload naturally being distributed a bit more to the upwards component as they create the output. However, it is also up to the member in the downstream component to adjust their components to the input.

9. GANTT CHART

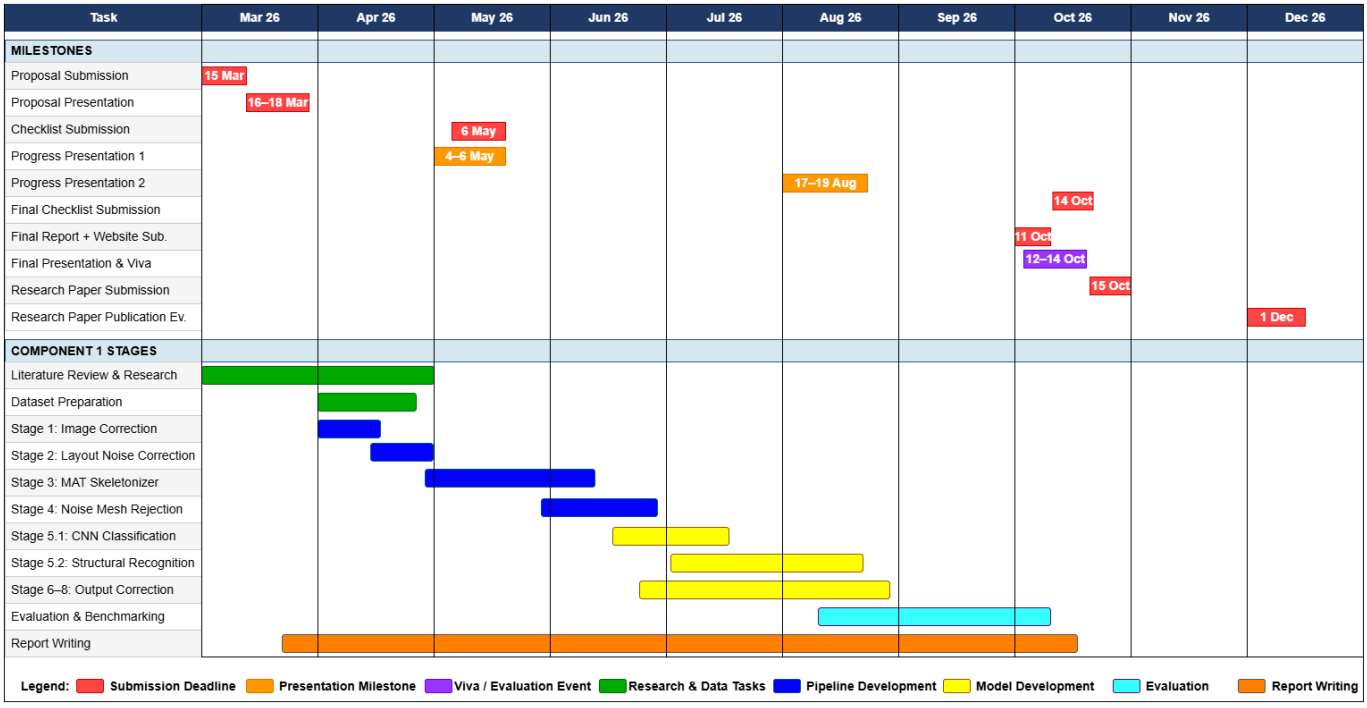


Figure 4:Gantt Chart

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11.APPENDICES

Risk Analysis

Table 4:Risk Analysis

Risk	Likelihood	Impact	Mitigation
Insufficient labelled Sinhala headline dataset	High	High	Supplement with synthetic data generation; collaborate with publishers, create dataset personally
MAT producing noisy skeletons on low-quality print scans	Medium	High	Tune noise isolation parameters; apply morphological cleaning pre-MAT
Tortuosity threshold misclassifying valid diacritics as noise	Medium	Medium	Empirically calibrate threshold on held-out validation set
Computational latency too high for real-time glasses use	Low	High	Scope component to headline-only; optimize with NumPy vectorization or simply focus on still images
Approach B (route matching) too complex for project timeline	Medium	Low	Deprioritize to secondary phase; Approach A sufficient for primary delivery.

Ethical Documents

This project does not involve direct human participants in a research capacity — while the final system is designed for use by visually impaired individuals, human involvement is limited to end-user interaction with the assistive interface and does not extend to data collection from persons, clinical observation, behavioral studies, or any form of sensitive human subject research. Therefore, this project does not require informed consent documentation or human ethics board clearance. The research is governed by SLIIT's Faculty of Computing undergraduate research ethics guidelines. A data ethics declaration confirming the use of publicly available and openly licensed datasets is maintained by the researcher. Ethical clearance from print media publishers will be sought prior to any public dataset release if required.